

Phase 3 – Project Design Phase

PROJECT: EduGenie – Google Gemini Powered Learning Assistant

OBJECTIVE

To design the overall architecture, modules, user interface, and data flow of EduGenie before development.

SYSTEM DESIGN

User → HTML/CSS Frontend → FastAPI Backend → AI Modules → Result displayed to User

MAIN MODULES

1. Explanation Module – uses LaMni-Flan-T5 for simplified concept explanations.
2. QnA Module – uses Gemini for academic/general question answering.
3. Quiz Module – generates structured quiz questions with options and correct answers.
4. Summary Module – summarizes long content into concise notes.
5. Learning Path Module – creates personalized learning recommendations.

API DESIGN

• /qa

- /explain
- /quiz
- /summarize
- /learn/recommendations

FRONTEND DESIGN

• Task selection dropdown

- Text input area
- Submit button
- Result display area
- Responsive CSS styling

PROCEDURE

1. Design the system architecture.
2. Divide the application into independent modules.
3. Map each module to an API endpoint.
4. Design the frontend input and result sections.
5. Decide how the frontend communicates with the FastAPI backend.
6. Finalize the project structure.